Gideon A. Afriyie

Mathematics | Quantitative & Data Research

Gettysburg, PA
Website
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Education

Gettysburg College Aug 2023 – May 2027 (expected)
B.A. in Mathematics *Gettysburg, PA*
» AMS F-GAP Scholar, Taylor Gaw '13 Endowed Fellowship, Seibert Project Award

Publications

Extreme Value Theory Analysis of Prime Gap Distributions: Statistical Analysis of Cramér's Conjecture and Light-Tailed Behavior 2026
› *Statistics & Probability Letters* (Elsevier) | **Sole author** | DOI: 10.1016/j.spl.2026.110898

Predicting Urban Air Quality Trends Using Machine Learning and Public Satellite/Sensor Data 2026
› R. Haque, B. Kebede, **G. A. Afriyie**, H. Yusuf. *Summer Proceedings 2026*, Ch. 16 | Link

Projects

Quantitative Risk & Portfolio Optimization GitHub
› Analyzed daily prices for 8 assets from 2020–2026, estimating returns, volatility, and correlations (VOO–VIXY: -0.754).
› Built minimum-variance, efficient-frontier, and tangency portfolios, using KKT conditions to enforce no short selling; in-sample tangency Sharpe ratio of 1.29 vs. 0.48 for minimum variance.

Experience

Under Canvas Inc. May 2025 – Jul 2025
Data Analytics Fellow *Remote*
» Python, Excel
› Analyzed a $1.6M compensation dataset, comparing take-home pay, overtime, bonuses, and hours worked across employees.
› Found that bonus payouts showed no clear relationship to hours or overtime and that bonus types were applied inconsistently; recommended standardizing eligibility and tying incentives more directly to effort.
› Presented findings that supported the organization's cost estimate and budget planning.

Gettysburg College, Advancing Science Aug 2026 – Present
Research Assistant *On-campus*
» Python, QGIS, SQL, TouchTerrain
› Developing tactile 3D topographic models of South Mountain for K–12 geoscience education.
› Integrated elevation, transportation, hydrography, and trail datasets through filtering, clipping, and reprojection.
› Designed and tested 3D-printed terrain prototypes featuring raised roads, recessed waterways, and the Appalachian Trail.

Programs & Training

Stanford University, Probability for AI Oct 2026 – Nov 2026
› Probabilistic foundations of AI through projects in statistical inference, ML, and model evaluation.

MIT/Harvard, IAIFI AI Summer School Aug 2026
› Symbolic regression, diffusion models, and simulation-based inference; AI hackathon.

Google, TRC May 2026
› Accepted into Google's TPU Research Cloud program for ML research, with access to Cloud TPUs.

Skills

Tools Python, SQL, Excel, LaTeX